Worksheet 5
Actuarial Corporate Finance

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I think the only worksheet problem assigned for homework was the third problem, but I will rewrite the first two just in case.

1 Problem 1

1.1 Part 1

The risk-free rate is represented by the rate on Treasury bills, which is 4.8% here; accordingly, the excess return on small stocks is $\boxed{0.232 - 0.048 = 18.4\%}$

1.2 Part 2

The excess return on corporate bonds is $\boxed{0.075 - 0.048 = 2.7\%}$

1.3 Part 3

The market risk premium is approximated by the S&P 500; accordingly, the excess return is $\boxed{0.132 - 0.048 = 8.4\%}$

1.4 Part 4

Using CAPM, we have

$$0.075 = 0.048 + \beta_{CB}(0.132 - 0.048) \implies \boxed{\beta_{CB} = 0.3214}$$

1.5 Part 5

The relationship between the excess return of the corporate bonds portfolio and the return of the market can be expressed in terms of volatility as

$$\mathbb{E}[RCB] = r_f + \frac{\sigma_{CB}}{\sigma_M}(\mathbb{E}[R_M] - r_f) \implies \boxed{\sigma_{CB} = \frac{0.084}{0.2 \times 0.027} = 6.43\%}$$

2 Problem 2

The current portfolio beta is given by $\beta_P = 0.6 \times 1.5 + 0.4 \times 1.25 = 1.4$. To lower it to 0.5, we need to use a different linear combination of assets. We can get this by writing

$$\beta_P = 0.5 = 1.4w + 0(1-w) \implies w = \frac{0.5}{1.4} = 0.3571$$

As written, w is the remaining weight on the portfolio, so the percentage invested in the risk-free bond should be $\boxed{\alpha = 1 - 0.3571 = 0.6429}$.

3 Problem 3

3.1 Part 1

The CAPM gives $\mathbb{E}[R_i] = r_f + \beta i, M(\mathbb{E}[R_M] - r_f)$. The only difference is $\beta_{i,M}$, which immediately tells us that the asset with a higher β should have a higher expected return. That means that asset A has a higher expected return.

3.2 Part 2

Plugging numbers into the CAPM, we get

$$\mathbb{E}[R_A] = 0.03 + 2.1(0.09 - 0.03) = 15.6\%$$
$$\mathbb{E}[R_B] = 0.03 + 0.7(0.09 - 0.03) = 7.2\%$$

3.3 Part 3

The portfolio's beta and expected return are given by

$$\mathbb{E}[R_{P1}] = 0.25\mathbb{E}[R_B] + 0.75\mathbb{E}[R_A] = 0.25 \times 0.072 + 0.75 \times 0.156 = 13.5\%$$
$$\beta_{P1} = 0.25 \times \beta_B + 0.75 \times \beta_A = 0.25 \times 0.7 + 0.75 \times 2.1 = 1.75$$

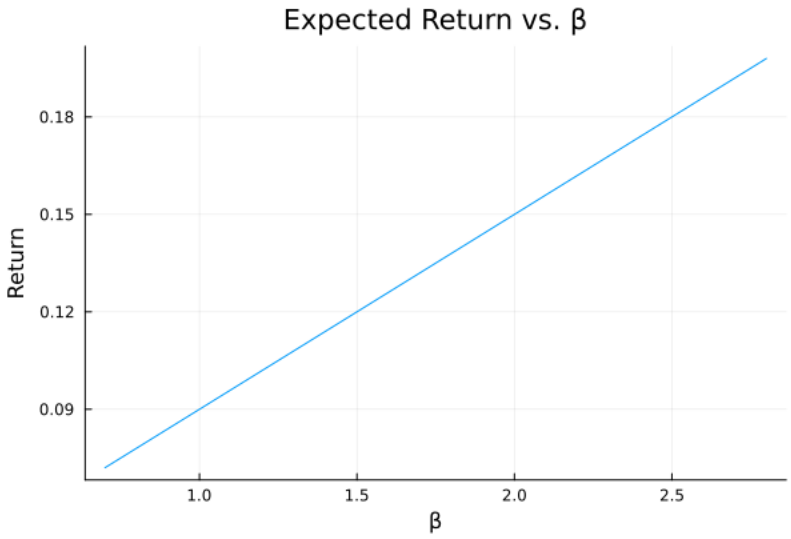
3.4 Part 4

This portfolio's beta and expected return are given by

$$\mathbb{E}[R_{P2}] = -0.5\mathbb{E}[R_B] + 1.5\mathbb{E}[R_A] = -0.5 \times 0.072 + 1.5 \times 0.156 = 19.8\%$$
$$\beta_{P2} = -0.5 \times \beta_B + 1.5 \times \beta_A = -0.5 \times 0.7 + 1.5 \times 2.1 = 2.8$$

3.5 Part 5

The assets clearly lie along a straight line. With a quick and dirty Julia script:




```
using Plots

R = [0.156, 0.072, 0.135, 0.198]
beta = [2.1, 0.7, 1.75, 2.8]

betarplot = plot(beta, R, xlabel = "beta", ylabel = "Return",
title = "Expected Return vs. beta", legend = false)
#definitely a straight line
png(betarplot, "ws5_plot")
```